

Directional response jets of suppressed multiphoton events in a four-mode Fourier interferometer

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Suppressed output events are established signatures of multiphoton interference, but a zero measured at one operating point does not reveal how the event responds to coherent implementation errors. We resolve this ambiguity for selected events of the four-mode Fourier interferometer by expanding their probabilities along a complete set of calibrated $SU(4)$ output directions. Depending on the event and direction, the probability remains identically zero or begins at quadratic or quartic order. For the four-photon transition $(0, 1, 2, 1) \rightarrow (0, 1, 2, 1)$, one beam-splitter quadrature obeys $P_{Y_{13}}(\epsilon) = 0$ for every angle, whereas the orthogonal quadrature obeys the exact law $P_{X_{13}}(\epsilon) = \sin^2(2\epsilon)/16$. The exact null extends to $(0, a, 2a, a) \rightarrow (0, a, 2a, a)$ for every positive odd a . As a structural tool, we derive an F_4 -specific, phase-resolved sectorwise reciprocity from classical binary Krawtchouk duality. It maps this apparently nonperiodic family to cyclic suppression and preserves reciprocal amplitudes throughout the standard dephased order-four complex-Hadamard family. We give all twelve off-diagonal $SU(4)$ response directions, an eleven-photon example with quartic onset, and a finite-shot nuisance-floor model. These event-level response jets refine disorder-averaged robustness estimates and provide falsifiable quadrature-sensitive checks for programmable photonic circuits.

I. INTRODUCTION

Exact zeros in multiparticle transition probabilities generalize the Hong–Ou–Mandel effect to multiport interferometers. Fourier suppression laws [1–4] and their extensions [5] are useful both as interference phenomena and as device diagnostics. Experiments have used forbidden-event populations to quantify departures from ideal behavior [2, 6], while recent protocols use Fourier-suppressed outputs to benchmark genuine multiphoton indistinguishability [7]. Partial distinguishability and generic weak disorder therefore have a mature theory [3, 8].

Here we ask a different question: *what is the coherent directional response of a specified dark event?* A random scalar disorder strength averages over physically distinct tangent directions. A programmable interferometer instead allows one to append a calibrated two-mode rotation and scan its signed angle. The resulting Taylor series—the event’s response jet—can contain directional information that is erased by averaging.

Our principal results are:

1. exact neighboring-permanent formulas give the response along all twelve off-diagonal generators of $SU(4)$, while the three diagonal directions preserve every dark event;
2. selected Fourier-dark events exhibit exact, quadratic, and quartic probability onsets, rather than one universal robustness exponent;

3. the four-photon event $\mathbf{r} = \mathbf{s} = (0, 1, 2, 1)$ has the exact quadrature contrast

$$P_{Y_{13}}(\epsilon) = 0, \quad P_{X_{13}}(\epsilon) = \frac{\sin^2(2\epsilon)}{16}; \quad (1)$$

4. an F_4 -specific sectorwise refinement of classical Krawtchouk/symmetric-power reciprocity [9–11] organizes zeros that evade a direct cyclic test.

We do not claim that the displayed response of one event is a universal invariant of every zero generated by the same broad suppression mechanism. The experimentally falsifiable objects here are *event–direction pairs*. This restriction is important: different members of a suppression family can have different stabilizers and different response jets.

II. FOURIER AMPLITUDES AND PHASE BINS

Let $\mathbf{r} = (r_0, \dots, r_3)$ and $\mathbf{s} = (s_0, \dots, s_3)$ be input and output occupations with $\sum_j r_j = \sum_j s_j = n$. For a single-particle unitary U , the Fock transition amplitude is

$$\mathcal{A}_{\mathbf{r},\mathbf{s}}(U) = \frac{\text{per } U[\mathbf{s}, \mathbf{r}]}{\sqrt{\prod_j r_j! \prod_k s_k!}}, \quad (2)$$

where rows and columns are repeated according to $\mathbf{s}$ and $\mathbf{r}$. We use the canonical dephased and ordered Fourier matrix

$$(F_4)_{jk} = \frac{i^{jk}}{2}, \quad j, k \in \{0, 1, 2, 3\}. \quad (3)$$

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Before the common factor 2^{-n} is inserted, each labelled permanent path has one of four phases. We write

$$\mathbf{h}(\mathbf{r}, \mathbf{s}) = (h_0, h_1, h_2, h_3), \quad \text{per}(2F_4)[\mathbf{s}, \mathbf{r}] = \sum_{\ell=0}^3 h_\ell i^\ell. \quad (4)$$

The histogram is a combinatorial object in this fixed gauge; it is not itself invariant under arbitrary row and column rephasings. Equality of transition probabilities and simultaneous zero/nonzero status are gauge invariant.

For reference, the elementary cyclic rule follows by shifting a periodic occupation. If the input is invariant under a shift τ , then a nonzero transition requires

$$\tau \sum_{j=0}^3 j s_j = 0 \pmod{4}, \quad (5)$$

with the reciprocal output condition following from the symmetry of F_4 [1, 4].

III. PHASE-RESOLVED SECTORWISE RECIPROCITY

Classical symmetric-power identities transpose all input and output indices for a symmetric matrix [10, 11]. The binary factorization of F_4 permits a partial exchange: the odd-mode split can be transposed while both even-mode splits are held fixed.

Sectorwise reciprocity theorem. Let $d, N \geq 0$, $0 \leq \alpha, s \leq d$, and $0 \leq k, p \leq N$. Define

$$\mathcal{H}_{d,N}(\alpha, k; s, p) = \mathbf{h}((\alpha, k, d - \alpha, N - k), (s, p, d - s, N - p)). \quad (6)$$

Then

$$\begin{aligned} \mathcal{H}(\alpha, k; s, p) &= \mathcal{H}(\alpha, p; s, k) \\ &= \mathcal{H}(s, k; \alpha, p) = \mathcal{H}(s, p; \alpha, k). \end{aligned} \quad (7)$$

Thus the complete four-bin phase count, not only the Fourier amplitude, is preserved. The visible hypothesis is that input and output have the same total occupation d in the even-indexed modes (and hence the same total N in the odd-indexed modes).

We give a compact proof. For $q^4 = 1$, let

$$\Phi_q(\mathbf{r}, \mathbf{s}) = \left(\prod_j r_j! \right) [\mathbf{x}^{\mathbf{r}}] \prod_{u=0}^3 \left(\sum_{v=0}^3 q^{uv} x_v \right)^{s_u}. \quad (8)$$

The four values $\Phi_1, \Phi_i, \Phi_{-1}, \Phi_{-i}$ are the four-point discrete Fourier transform of $\mathbf{h}$. At $q = i$, introduce

$$E = x_0 + x_2, \quad T = x_0 - x_2, \quad S = x_1 + x_3, \quad D = x_1 - x_3. \quad (9)$$

The four forms become $E + S, T + iD, E - S, T - iD$. After expanding the even sector, every remaining summand contains

$$K_m(p; N) K_k(m; N), \quad (10)$$

where

$$K_m(p; N) = \sum_t (-1)^t \binom{p}{t} \binom{N-p}{m-t} \quad (11)$$

is a binary Krawtchouk coefficient. Self-duality gives

$$\binom{N}{p} K_m(p; N) = \binom{N}{m} K_p(m; N). \quad (12)$$

Applied twice, Eq. (12) exchanges p and k term by term. The binomial ratio is canceled exactly by the $k!(N-k)!$ and $p!(N-p)!$ factors in Eq. (8). At $q = -1$, the polynomial is $(E + S)^d (E - S)^N$; the requested total degree d in (x_0, x_2) forces the $E^d S^N$ contribution, and the same factorial cancellation applies. The $q = 1$ value counts all labelled paths, and $q = -i$ follows by conjugation. Fourier inversion proves Eq. (7). Details are in the Supplemental Material.

The operation is generically not a mode relabelling. For example, with $d = 5$, $N = 11$, $(\alpha, s, p, k) = (1, 0, 3, 2)$, the input occupation multisets on the two sides are $\{1, 2, 4, 9\}$ and $\{1, 3, 4, 8\}$, while the output multisets are $\{0, 3, 5, 8\}$ and $\{0, 2, 5, 9\}$. They remain unequal after global input-output exchange.

A. Polynomial reciprocity for the order-four Hadamard family

Consider

$$H(z) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & z & -1 & -z \\ 1 & -1 & 1 & -1 \\ 1 & -z & -1 & z \end{pmatrix}. \quad (13)$$

Replacing i by z in the preceding coefficient proof adds a common factor z^{N-m} . The reciprocal permanents are therefore equal as polynomials in z . When $|z| = 1$, $H(z)/2$ is the standard dephased one-parameter order-four complex-Hadamard family [12], and the normalized Fock amplitudes are equal. This is amplitude reciprocity, not persistence of every Fourier zero: already for $\mathbf{r} = \mathbf{s} = (0, 1, 2, 1)$ the relevant unnormalized coefficient is proportional to $2(1 + z^2)$ and is generically nonzero.

B. Closure of a nominally noncyclic family

For

$$\mathbf{r} = \mathbf{s} = (0, a, 2a, a), \quad (14)$$

rotations and Eq. (7) transport the zero condition to

$$(0, 0, 2a, 2a) \longrightarrow (a, a, a, a). \quad (15)$$

The latter is suppressed by Eq. (5) precisely when a is odd. Thus Eq. (14) is not evidence for a separate selection mechanism; it lies in the reciprocity closure of a cyclic law. An exact enumeration illustrates the compression produced by this closure:

n	residual reps.	closure components	reach cyclic
4	3	3	1
5	8	3	0
6	10	6	0
7	0	0	0
8	33	23	1
9	72	40	0

These finite values are an application of the theorem, not an asymptotic classification claim.

IV. DIRECTIONAL RESPONSE JETS

Append a calibrated two-mode rotation after F_4 :

$$\begin{aligned} U_{pq}^X(\epsilon) &= e^{i\epsilon X_{pq} F_4}, & X_{pq} &= |p\rangle\langle q| + |q\rangle\langle p|, \\ U_{pq}^Y(\epsilon) &= e^{i\epsilon Y_{pq} F_4}, & Y_{pq} &= -i|p\rangle\langle q| + i|q\rangle\langle p|. \end{aligned} \quad (16)$$

The six unordered pairs and two quadratures span the twelve off-diagonal directions of $\mathfrak{su}(4)$. The remaining three traceless diagonal generators append output phases; they multiply a fixed Fock amplitude by a phase and therefore preserve every exact zero.

Let $Z_{\mathbf{r}, \mathbf{t}} = \text{per}(2F_4)[\mathbf{t}, \mathbf{r}]$ and

$$D_{\mathbf{r}, \mathbf{s}} = 2^n \sqrt{\prod_j r_j! \prod_k s_k!}. \quad (17)$$

For a dark target, Fock ladder factors give the exact tangents

$$\mathcal{A}'_X(0) = \frac{i}{D_{\mathbf{r}, \mathbf{s}}} (s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q} + s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q}), \quad (18)$$

$$\mathcal{A}'_Y(0) = \frac{1}{D_{\mathbf{r}, \mathbf{s}}} (s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q} - s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q}). \quad (19)$$

Negative occupations are omitted. If the tangent is nonzero, $P(\epsilon) = |\mathcal{A}'(0)|^2 \epsilon^2 + O(\epsilon^3)$. If it vanishes, Krylov moments of the two-mode generator decide whether a higher-order onset occurs or the amplitude is identically zero.

A. Four-photon benchmark predictions

Table I gives the complete off-diagonal response for two specified four-photon dark events. “Exact” means zero for every rotation angle on that axis.

For the cyclic example, arbitrary $U(2)$ rotations on pairs 02, 12, 13, 23 preserve the zero. Each rotation preserves the total occupation in its pair, and none of the reachable modular sums satisfies Eq. (5). For the second example, direct coefficient evaluation strengthens the small-angle 13 entries to Eq. (1). The two quadratures therefore give a clean same-pair contrast.

The Y_{13} null is not a four-photon accident. For every positive odd a , the event in Eq. (14) remains dark under the full Y_{13} rotation. With $x = x_1$, $y = x_2$, $z = x_3$, $u = x + z$, and $v = x - z$, its coefficient is proportional to

$$[x^a y^{2a} z^a] (A(y^2 + v^2) + Byv)^a (y - u)^{2a}. \quad (20)$$

Reflection $x \leftrightarrow z$ removes odd powers of v . After choosing $2h$ factors of Byv , the remaining sum pairs an index k with $a - k$. Their weights agree, whereas

$$\begin{aligned} [x^a z^a] (x - z)^{2(a-k)} (x + z)^{2k} \\ = (-1)^a [x^a z^a] (x - z)^{2k} (x + z)^{2(a-k)}. \end{aligned} \quad (21)$$

Every pair cancels for odd a , for every ϵ .

B. A quartic direction

Higher-order flatness need not imply exact protection. For

$$\mathbf{r} = (0, 1, 3, 7), \quad \mathbf{s} = (1, 3, 3, 4), \quad (22)$$

an integral zero on a previously scanned affine family, the Y_{12} tangent vanishes but the second derivative does not:

$$P_{\mathbf{r}, \mathbf{s}}^{Y_{12}}(\epsilon) = \frac{315}{8192} \epsilon^4 + O(\epsilon^5). \quad (23)$$

The other eleven off-diagonal directions and the exact Gaussian-integer certificate are given in the Supplemental Material. Equation (23) is a local quartic onset, not

TABLE I. Leading ideal probabilities for two four-photon events.

event	pair	X_{pq}	Y_{pq}
cyclic	01	$3\epsilon^2/8$	$3\epsilon^2/8$
$\mathbf{r} = (1, 1, 1, 1)$	02	exact	exact
$\mathbf{s} = (3, 1, 0, 0)$	03	$3\epsilon^2/8$	$3\epsilon^2/8$
	12	exact	exact
	13	exact	exact
	23	exact	exact
reciprocity-closed	01	$\epsilon^2/64$	$\epsilon^2/64$
$\mathbf{r} = \mathbf{s} =$	02	$\epsilon^2/4$	$\epsilon^2/4$
$(0, 1, 2, 1)$	03	$\epsilon^2/64$	$\epsilon^2/64$
	12	$\epsilon^2/64$	$25\epsilon^2/64$
	13	$\epsilon^2/4$	exact
	23	$\epsilon^2/64$	$25\epsilon^2/64$

an all-angle null and not a claim that the event is globally isolated among all occupations.

V. APPLICATIONS AND FINITE-SHOT PREDICTIONS

Quadrature-sensitive calibration. Equation (1) predicts the entire ideal angle dependence for two orthogonal quadratures on the same pair of modes. Counts on the Y_{13} axis reveal departure from the joint ideal model, while the X_{13} curve supplies an in situ scale for the programmed angle. This does not replace full unitary tomography [13]; it is an event-level, multiphoton consistency check.

Cross-configuration consistency. Equation (7) produces occupation pairs with equal ideal probabilities for every $|z| = 1$ in Eq. (13). Sweeping z and comparing reciprocal configurations overdetermines the ideal prediction. A discrepancy diagnoses failure of the joint state–interferometer–detector model, but by itself does not identify which component failed.

Protected analysis subgroups. The exact rows of Table I identify output-mode subgroups within which a chosen target remains dark. They can be used to modify the analysis basis without opening that event. This is an event-specific design constraint, not a fault-tolerant subspace.

Separating a floor from coherent response. To quantify a near-term experiment, we use the transparent nuisance model

$$P_{\text{obs}}(\epsilon) = \mathcal{V}P_{\text{ind}}(\epsilon) + (1 - \mathcal{V})P_{\text{dist}}(\epsilon) + B, \quad (24)$$

where $\mathcal{V}$ is a run-level mixture weight, P_{dist} is the fully distinguishable labelled-photon probability in the same network, and B is an additive accepted-trial background. Equation (24) is not asserted to be a universal partial-distinguishability model; general Gram-matrix treatments are given in Refs. [8, 14].

At $\epsilon = 0.1$, $\mathcal{V} = 0.99$, and $B = 0$, the $(0, 1, 2, 1)$ target has

$$\begin{aligned} P_{\text{obs}}^{X_{13}} &= 2.91093 \times 10^{-3}, \\ P_{\text{obs}}^{Y_{13}} &= 4.74917 \times 10^{-4}. \end{aligned} \quad (25)$$

The exact ideal contributions are 2.46684×10^{-3} and zero. Under independent binomial sampling, the normal approximation for the difference of two proportions gives

$$M \simeq \frac{Z^2[P_X(1 - P_X) + P_Y(1 - P_Y)]}{(P_X - P_Y)^2}, \quad (26)$$

or $M = 14228$ accepted four-photon trials per setting for $Z = 5$. This number is illustrative, model-dependent, and reproducible from the released script; it is not a hardware-independent sample-complexity theorem.

The same model makes several failure modes explicit. A small axis misalignment δ away from Y_{13} toward X_{13}

produces the leading ideal term

$$P(\epsilon, \delta) = \frac{\epsilon^2 \sin^2 \delta}{4} + o(\epsilon^2), \quad (27)$$

so the null is a sensitive quadrature-alignment check. Pure attenuation after the interferometer multiplies a fixed target amplitude and cannot lift an exact zero when conditioning on all n detected photons. By contrast, internal or path-dependent loss can alter relative paths; source contamination and detector noise can add a floor. These contributions must be independently characterized before counts on a null axis are assigned to one cause.

VI. RELATION TO PRIOR ROBUSTNESS RESULTS

Dittel *et al.* derived generic quadratic leakage under weak random unitary disorder and linear leakage in small distinguishability parameters after disorder averaging [3]. Our result is compatible with that scaling but retains the coherent generator before averaging. The response may vanish on a tangent direction, begin at fourth order, or remain zero along an entire one-parameter subgroup. Averaging over generic directions suppresses precisely this anisotropic information.

Our four-bin histogram should also not be confused with the Fourier transform over the permutation group used to resolve exchange-symmetry sectors [14]. Here the bins count root-of-unity phases of permanent paths in a fixed F_4 gauge. The Krawtchouk step is a sectorwise, phase-resolved application of mature symmetric-power machinery, not a new Krawtchouk identity.

VII. DISCUSSION

The main conceptual lesson is that “robustness of a suppressed event” is not a scalar without a perturbation model. Selected F_4 events have distinct response jets across experimentally addressable mixer quadratures. The cleanest near-term test uses four photons and the exact X_{13}/Y_{13} contrast in Eq. (1). The eleven-photon quartic direction is a higher-resource prediction.

Several limitations define the next work. A hardware claim requires a reconstructed circuit and a full distinguishability matrix rather than the mixture in Eq. (24). Internal component errors must be pushed through the circuit to determine which output tangent directions they excite. The finite closure census does not classify all Fourier zeros, and the naive parity-sector analogue already fails for F_8 . Finally, the sectorwise reciprocity lies close to classical multivariate Krawtchouk theory; the priority claim made here is limited to its F_4 phase-resolved use and the coherent response consequences.

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DATA AND CODE AVAILABILITY

All data are generated by the source code accompanying this manuscript. The exact phase-histogram,

reciprocity-census, leakage, and finite-shot scripts use only the Python standard library. A permanent public repository is maintained at <https://github.com/HainanZhao/maph>; the submission archive also contains the complete code and tests needed for the reported results.

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